

Managing Log Files with the Logrotate Utility

Log files, though useful to troubleshoot and to track usage, tend to use up valuable disk space. Over time, they become large and unwieldy, so pinpointing an event becomes difficult. Logrotate performs the function of archiving a log file and starting a new one, thereby 'rotating' it.



Logrotate has been designed to ease the administration of systems that generate large numbers of log files in any format. It allows automatic rotation, compression, removal and mailing of log files. Each log file may be handled daily, every week, every month, or when it grows too large (rotation on the basis of a file's size).

The application and the servers generate too many logs, making the task of troubleshooting or gaining business insights from these logs, a difficult one. Many a time, there's the issue of servers running on low disk space because of the very large log files on them.

Servers with huge log files create problems when the resizing of virtual machines needs to be done. Troubleshooting based on large files may take up a lot of time and valuable memory. The logrotate utility is extremely useful to solve all such problems. It helps in taking backups of log files on an hourly, daily, weekly, monthly or yearly basis with additional choice of log backup with compression. Also, file backups can be taken by setting a limit on the file size, like 100MB, for instance. So, after the log file reaches a size of 100MB, the file will be rotated.

The synopsis is as follows:

```
logrotate [-dv] [-f|--force] [-s|--state file] config_file
```

Any number of configuration files can be given on the command line, and one file can include another config file. A simple logrotate configuration looks like what's shown below:

```
/var/log/messages {
    rotate 5
    weekly
    compress
    olddir /var/log/backup/messages/
    missingok
}
```

Here, every week, the `/var/log/messages` file will be compressed and backed up to the `/var/log/backup/messages/` folder, and only five rotated log files will be kept around in the system.

Installing logrotate

Log rotation is a utility that comes preinstalled in Linux servers like Ubuntu, CentOS, Red Hat, etc. Check the folder at path `/etc/logrotate.d`. If it is not installed, then you can install it manually by using the following commands.

For Ubuntu, type:

```
sudo apt-get install logrotate
```